

DevLearn - Learning Platform

Project Documentation

1. Project Overview

DevLearn is a modern, full-featured learning management system (LMS) for web development. It provides interactive courses with lessons, flashcards, quizzes, video sessions, articles, progress tracking, and user authentication.

2. Tech Stack

Next.js 16.2.7 (App Router) - React framework with server components

Tailwind CSS v4 - Utility-first CSS framework

SQLite / PostgreSQL (Neon) - Database (local / production)

Prisma 7.8.0 - ORM with driver adapter pattern

NextAuth v5 - Authentication (GitHub, Google, Email)

next-mdx-remote - MDX rendering

bcryptjs - Password hashing

TypeScript - Type safety

3. Authentication

Three sign-in methods are supported:

- GitHub OAuth
- Google OAuth
- Email & Password (bcrypt hashed)

Account linking is supported across providers using the same email.

4. Database (10 Models)

User, Account, Session, VerificationToken, Course, Lesson, Flashcard, QuizQuestion, VideoSession, UserProgress

5. Courses (5 Courses, 26 Lessons)

Course	Lessons	Videos
HTML Basics	5 lessons	2 video sessions
CSS Fundamentals	5 lessons	2 video sessions
JavaScript Basics	5 lessons	2 video sessions
Python Basics	6 lessons	3 video sessions
C++ Basics	4 lessons	1 video session

6. Articles (6 Articles)

Comprehensive guides for HTML, CSS, JavaScript, Python, Python OOP, and C++.

7. Routes (15 Total)

/	Landing page
/courses	Course listing
/courses/[slug]	Course detail
/courses/[slug]/[lesson]	Lesson viewer
/courses/[slug]/[lesson]/flashcards	Flashcards
/courses/[slug]/[lesson]/quiz	Quiz
/courses/[slug]/sessions/[sessionId]	Video player
/dashboard	Progress dashboard
/articles	Articles listing

<code>/articles/[slug]</code>	Article detail
<code>/login</code>	Sign in
<code>/register</code>	Create account
<code>/api/auth/[...nextauth]</code>	Auth API
<code>/api/progress</code>	Progress API

8. Features

- Landing page with gradient hero and animations
- Course browsing with progress indicators
- MDX-rendered lesson content with navigation
- Interactive flip flashcards with keyboard navigation
- Multiple-choice quizzes with instant scoring
- Embedded YouTube video sessions
- Dashboard with progress ring, stats, and activity feed
- Class-based dark mode with localStorage persistence
- Articles section with MDX rendering
- Authentication with 3 methods (GitHub, Google, Email)

9. Deployment (Vercel)

1. Push code to GitHub repository
2. Import repo at vercel.com/new
3. Add Vercel Postgres integration
4. Build command: `prisma db push && next build`
5. Post-install: `prisma generate`
6. Run seed script post-deployment

10. Local Development

```
npm install
npm run seed
npm run dev
```